

Orbital Oscillation

Tobias Klein*

School of Creative Media, City University of Hong Kong
Hong Kong, Hong Kong
ktobias@cityu.edu.hk

Pok Yin Victor Leung*

City University of Hong Kong
Hong Kong, Hong Kong
victor.leung@cityu.edu.hk

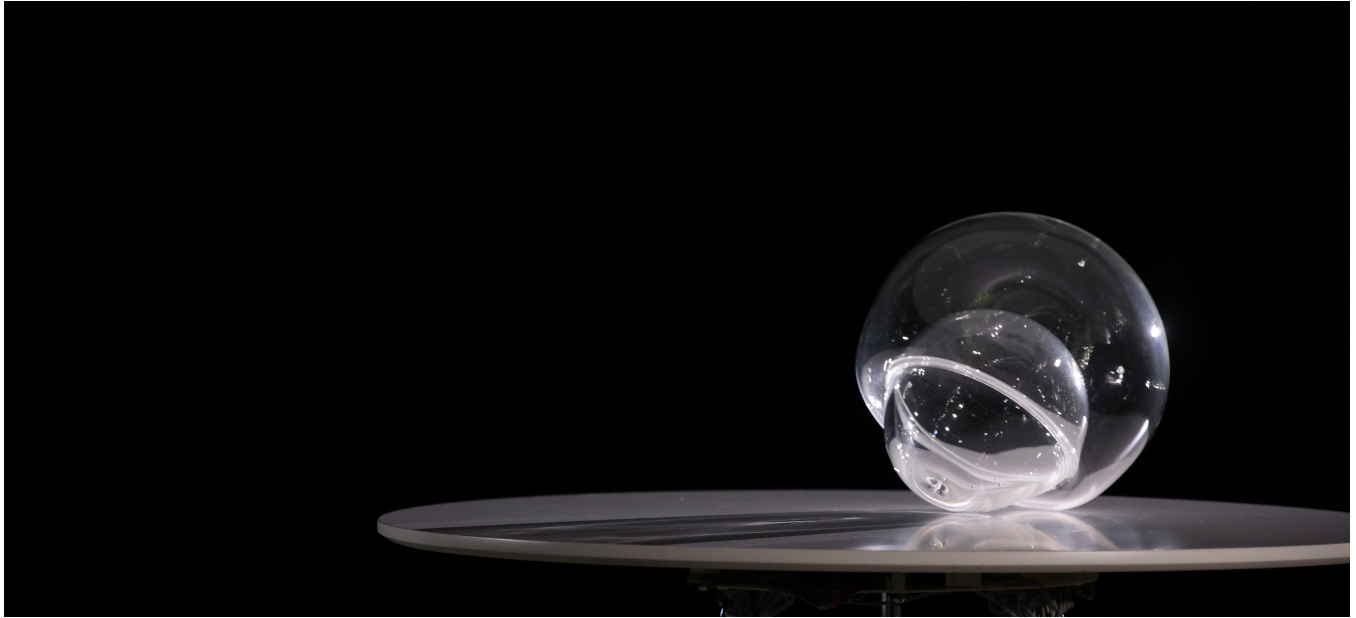


Figure 1: Detail image of the artwork *Orbital Oscillation*, all image credits are with the authors

Abstract

Orbital Oscillation is a kinetic sculpture exploring the phenomenology of time through the imperceptible actuation of a hand-blown glass object. The work presents a glass sphere rolling in slow, deliberate orbits on a minimalist table, its motion appearing autonomous yet precisely orchestrated by hidden robotic mechanisms beneath the surface. This concealment of technical agency—achieved through silent tilting actuators, real-time feedback control, and load-cell sensing—creates a perceptual tension: viewers witness motion without causality, inviting reflection on the relationship between visibility, agency, and expectation.

Conceptually, the work engages Husserl's model of internal time consciousness (retention, impression, protention), using cyclical motion and subtle disruptions to materialize the viewer's anticipatory experience. By destabilizing rhythmic predictions—accelerating,

reversing, or pausing the glass's trajectory—the sculpture foregrounds the fragility of perception itself. Philosophically, it resists Heidegger's *Gestell*, rejecting technological transparency in favor of *techne* as poetic disclosure: the mechanism remains invisible, allowing the object to "self-move" while quietly embodying Yuk Hui's *cosmotechnics* through planetary-like rhythms.

This work extends our earlier investigations in *Digital Craft*, where digital tools mediated material memory (e.g., 3D-printed scaffolds preserving glass fragments). Here, craft dematerializes: computation shifts from shaping physical form to choreographing temporal behavior. The glass, once constrained by visible structures, now rolls freely within an invisible technological framework—a transition from fabrication to temporal poetics. Engineering serves artistry not through spectacle but through restraint, rendering machine intelligence as a silent collaborator in an unfolding meditation on time, presence, and the unseen infrastructures shaping both.

Orbital Oscillation is a kinetic sculpture where a glass object rolls autonomously on a table, actuated by hidden mechanisms. Concealed tilting systems and real-time sensing create imperceptible motion, exploring time perception (Husserl) and technological poetics (Heidegger). The work dematerializes craft, shifting from physical structures to choreographed temporal experience.

Keywords

kinetic sculpture, Digital Craft, glass art, mechatronic design, feedback control, imperceptible motion, phenomenology

*Both authors contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Art Gallery '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1883-0/25/12

<https://doi.org/10.1145/3757368.3765546>



Figure 2: Image of the artwork Orbital Oscillation, all image credits are with the authors

ACM Reference Format:

Tobias Klein and Pok Yin Victor Leung. 2025. Orbital Oscillation. In *SIGGRAPH Asia 2025 Art Gallery (SA Art Gallery '25), December 15–18, 2025, Hong Kong, Hong Kong*. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757368.3765546>

1 Artist Statement

Our work explores the intersection of materiality, temporality, and perception through the evolving discipline of Digital Craft. By integrating traditional craft techniques with contemporary digital processes, we create objects and experiences that challenge the boundaries between the physical and the immaterial, the visible and the concealed.

In *Orbital Oscillation*, a hand-blown glass object performs deliberate orbits, animated by an imperceptible actuation system hidden within the table. This kinetic sculpture invites viewers into a dialogue between what is seen and what is caused, blending sensory ambiguity with poetic introspection. Drawing from phenomenological frameworks, particularly Edmund Husserl's notions of internal time consciousness, the work unfolds as a layered perceptual event where memory, presence, and expectation are constantly renegotiated.

Our practice reflects a deep engagement with the hidden agency of technology. Inspired by Martin Heidegger's notion of *techne* as a mode of revealing, we aim to render the technical process invisible, allowing focus to shift to the interplay of motion, rhythm, and anticipation. By concealing mechanisms and emphasizing the object's apparent autonomy, works like *Orbital Oscillation* challenge perceptions of causality, creating moments of quiet yet profound disruption.

From the crystalline forms of *Glass Entanglement* to the transient choreography of rolling glass, our work considers how material memory, movement, and digital mediation coexist. Ultimately, we strive to create spaces where the poetic and the technical merge, offering reflective experiences that reveal not just the inner workings of the artwork, but the viewer's own perceptions.

Acknowledgments

The authors gratefully acknowledge the support of the School of Creative Media, City University of Hong Kong. Special thanks to Mr. Zeyang Wu for his invaluable assistance with fabrication and documentation. This project was partially supported by the General Research Fund (CityU 11613124), under the project titled: Hybrid Glass Craft - Novel Approaches to Materials and Digital Tools in Glass Craft as a Model for the Synthesis of Art and Technology.